

École Centrale de Nantes

INFO SI Option – UI/UX Design and Mobile Dev (DEVMO)

Project: Coup de Feu

Interactive Assistance and "Hands-free" Kitchen Interface



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March 16, 2026

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I Introduction and Context

The **Coup de Feu** project addresses a major ergonomic paradox in professional kitchens: while digital data (order slips, timers, allergy alerts) are vital for performance, traditional tactile interaction is restricted by hygiene standards and physical motor constraints.

Our Android application enables fluid synchronization of the kitchen brigade through a highly reactive interface, filtering information by station and simulating touchless interaction modalities to preserve the chefs' workflow.

II UX Analysis and Design

1 Personas and User Journeys

Our design focused on two primary user profiles:

- **Sofia (Chef de Partie):** Works at the Fish station. She requires a "Micro" view of her specific tickets.
- **Marc (Head Chef):** Positioned at the "Pass." He needs a "Macro" view of the overall progress of each station (Cold, Fish, Meat) for every table.

2 Applied Ergonomic Criteria

1. **Guidance:** Use of exact colors from the Figma prototype (Orange for action, Blue for technical information, Red for urgency).
2. **Workload:** The system automatically calculates table progress. Sofia only sees tables requiring unfinished fish preparation.
3. **Explicit Control:** Implementation of a massive confirmation overlay ("GESTURE DETECTED" or "VALIDATED") with oversized icons (120sp).

III Technical Architecture

1 Software Stack

The application was developed entirely using modern Android technologies:

- **Jetpack Compose:** 100% of the user interface is declarative, with no XML layouts.
- **MVVM Architecture:** Use of a single CoupDeFeuViewModel shared across screens to ensure data consistency.
- **Reactive Data Flow:** Utilizing `StateFlow` and `collectAsState()` to update the UI in real-time when a dish is validated.

2 State Management and Persistence

To ensure robustness against Android lifecycle events, we implemented:

- **SavedStateHandle:** Serialization of Orders as delimited strings to restore the exact kitchen state after an app process death.
- **Derived State:** Sofia's ticket list is calculated dynamically from the global source of truth, filtering tables where the "fish" station is not yet complete.

IV Feature Implementation

1 Station View (Sofia) - Fish Station

Sofia's interface offers a streamlined dashboard featuring:

- A sidebar indicating station statistics (In progress, Alerts).
- A dynamic ticket grid. Each ticket displays the table number (56sp), items generated by a deterministic algorithm, and elapsed time.
- Two validation buttons simulating touchless interaction (Hand and Check icons).

2 Chef View (Marc) - Global Coordination

Marc's dashboard displays all 8 active tables with:

- Performance indicators (Active tables, Average service time).
- Three progress gauges per table (Cold, Fish, Meat) whose colors change dynamically based on completion rates ($< 50\%$ blue, $50 - 79\%$ orange, $\geq 80\%$ red).

3 Real-Time Synchronization

When a cook validates a preparation on the Station view, the ViewModel increments the corresponding station counter. The Chef view is instantly updated (gauge progress), illustrating fluid communication between brigade members.

4 Final Application Visualization

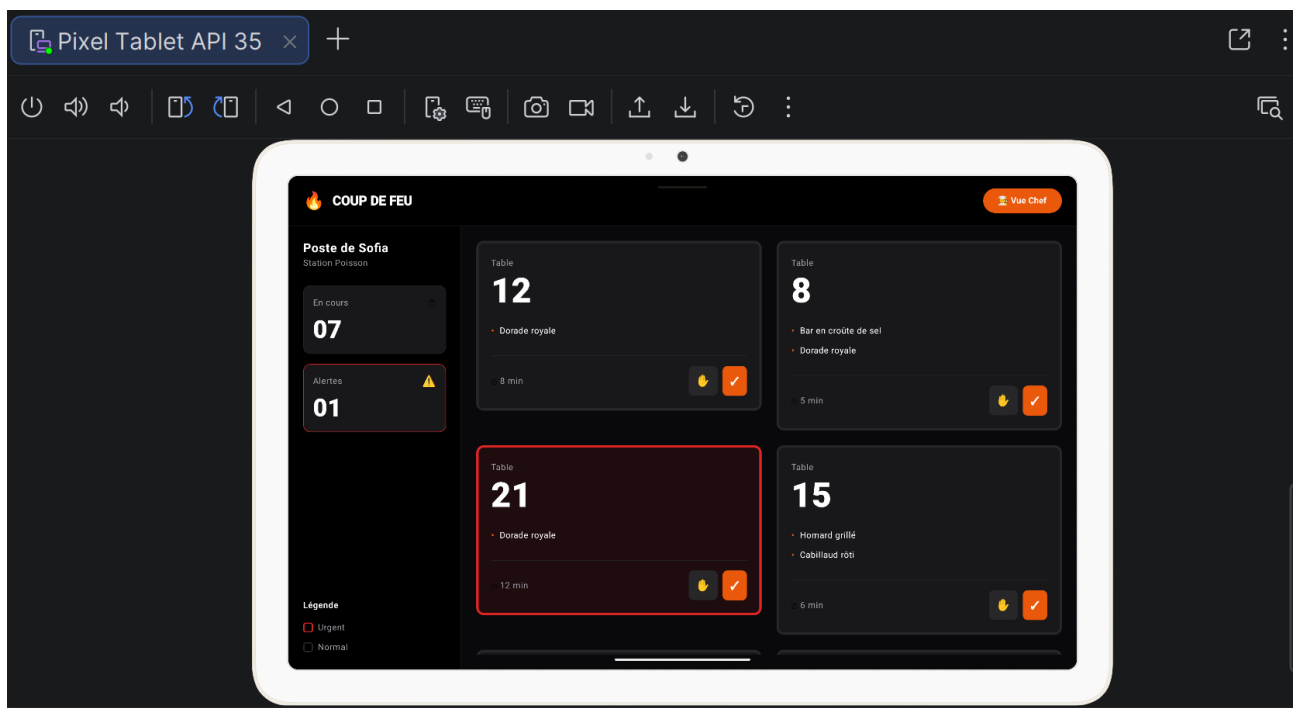


Figure 1: Micro-management interface designed for the Fish Station.

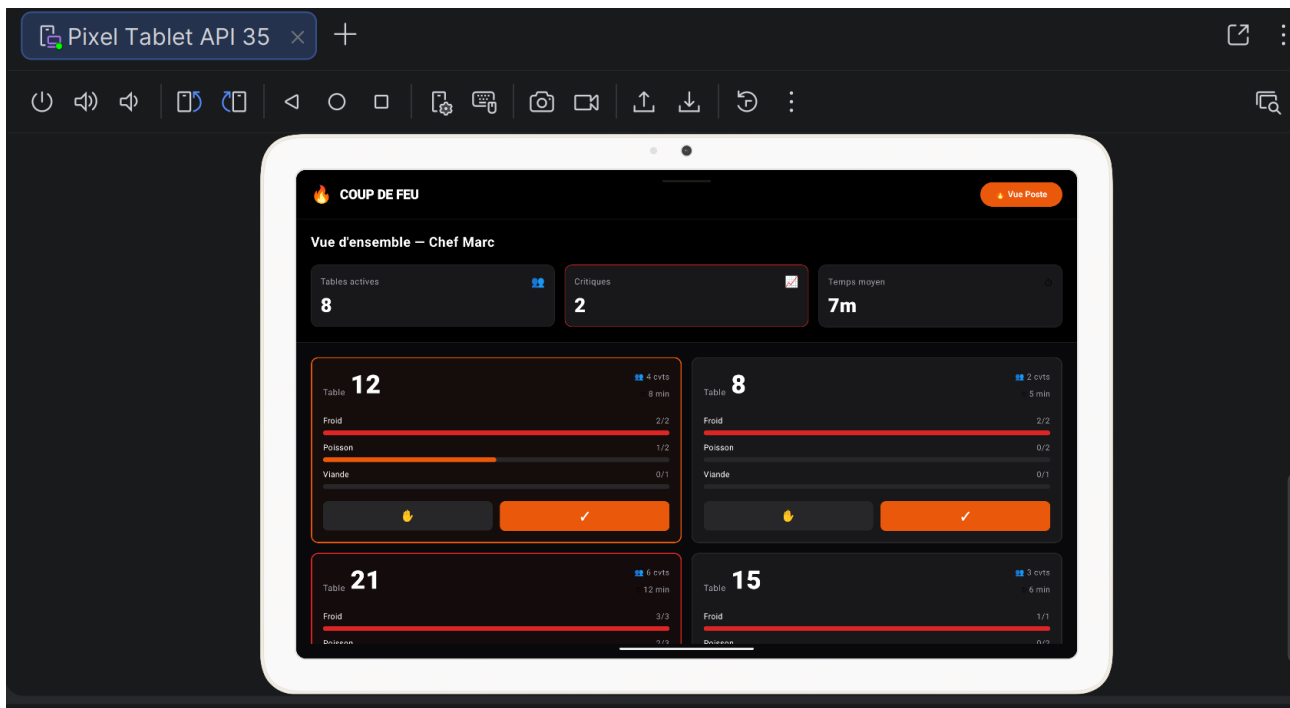


Figure 2: Global coordination interface for the Head Chef.

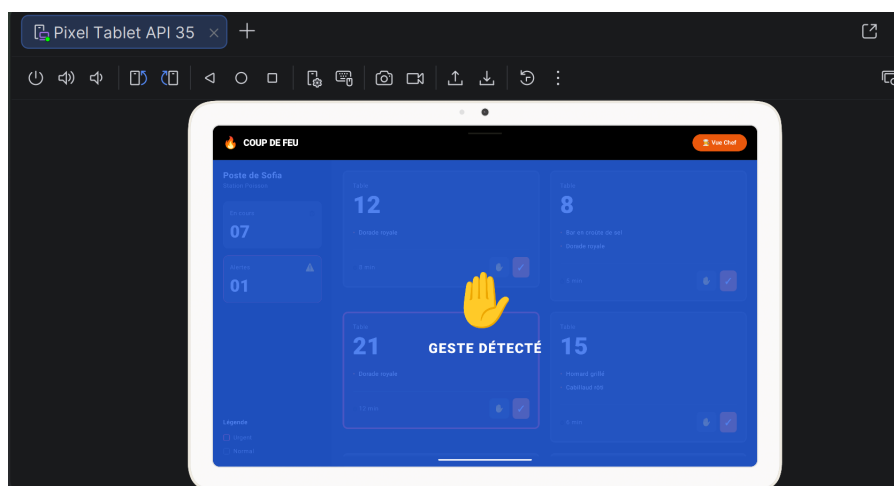


Figure 3: Massive visual confirmation upon touchless interaction detection.

V Conclusion and Future Outlook

The final implementation of **Coup de Feu** validates the ergonomic choices made at the start of the project. The use of Jetpack Compose allowed for the creation of a fluid and high-contrast interface, perfectly suited to the visual constraints of a kitchen environment. The source code is available on the team's Git repository, modularly structured (Models, ViewModel, UI, Theme).